Year 9 - 2015

<u>Investigation 1</u>

Semester One Term 1

Probability

Section 1: 0 Marks
Section 2: 12 Marks
Section 3: 18 Marks

This investigation has three sections. Section 1 will be completed in class as a group. Section 2 is to be completed at home and Section 3 will be completed in class under test conditions. For Sections 1 and 2 you are allowed to ask for help and guidance, however, Section 3 will be completed under test conditions to fully test your understanding. No guidance or hints will be provided.

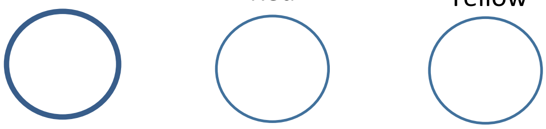
AIM: *In this assessment you will be investigating probabilities of two step events, calculating relative frequencies both with and without replacement. This will be done using both diagrams and calculations.*

<u>Section 1</u>

Name: ____ Class: ___________________

You have a bag which contains 3 coloured balls: one red, one blue and one yellow.

Blue Red Yellow



$$Theoretical\ probability = \frac{the\ \ number\ of\ favourble\ outcomes}{the\ number\ of\ possible\ outcomes}$$

1. Use the formula above to work out the following theoretical probabilities:

   a. Pr (red)

   b. Pr (blue)

   c. Pr (not yellow)

d. Pr (white)

2. As a class we will now choose 2 balls, replacing the first ball before choosing the second. Record the results in the table below:

First	Second	Tally	Frequency	Relative Frequency
Red	Red			
Red	Blue			
Red	Yellow			
Blue	Red			
Blue	Blue			
Blue	Yellow			
Yellow	Red			
Yellow	Blue			
Yellow	Yellow			
		Total		

Use the table to answer the following questions:

a. Pr (red then blue)

b. Pr (yellow then yellow)

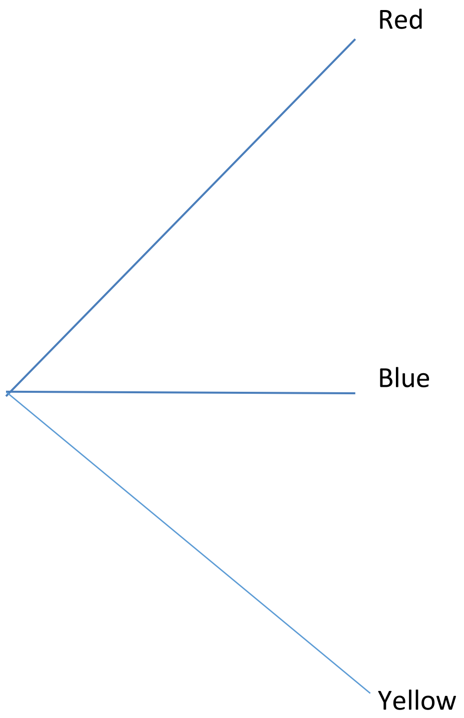
c. Pr (blue and yellow in any order)

d. Pr (red and blue and any order)

e. Pr (at least one ball red)

3. Draw a tree diagram to show the theoretical probability of the experiment carried out. The first branch has been drawn for you.

First choice | Second choice | Probability



How do the theoretical probability and experimental probability (relative frequency) compare?

Section 2 (Take Home Section)

Mark: /12

You will complete the experiment as we did in class, but your experiment should be without replacement. The aim of the experiment is to compare theoretical and experimental probability.

1. How many times do you think you should complete the experiment? Give reasons for your answer.

(2 marks)

2. Complete a frequency table to record your results:

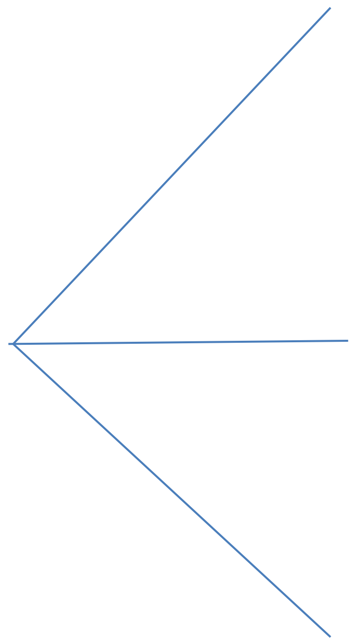
(4 marks)

(The following table has been provided as a suggestion. Not all rows may be required.)

First	Second	Tally	Frequency	Relative Frequency
Blue	Red			
Blue	Yellow			
		Total		

3. Draw a tree diagram to demonstrate the theoretical probability for this experiment.

(4 marks)



4. Is there a difference between the theoretical and relative frequency (experimental probability) for the outcomes? Explain why/why not.

(2 marks)